2.1 Conditional Statements

Name _____ Period ____ Date _____

Geometry 2.1 (30-minute plan) · Student sheet

I can…

- identify the hypothesis and conclusion of a conditional statement.
- write the converse, inverse, and contrapositive and decide whether each is true.
- write a biconditional statement.

Vocabulary · fill in the missing word

Conditional statement — a logical statement with a _____ and a conclusion, written "if p, then q"

Converse — the statement formed by switching the _____ and the conclusion

Inverse — the statement formed by negating both the _____ and the conclusion

Contrapositive — the statement formed by switching and negating the _____ and the conclusion

Biconditional statement — a statement that _____ the phrase "if and only if"

Part 1 · Notes and examples (fill in each note, then work the example)

Conditional statement: "if p, then q"; p is the _____ and q is the _____.

Example 1

Identify the hypothesis and the conclusion: "If a polygon has 5 sides, then it is a pentagon."

Related conditionals: converse $q \to p$, inverse $\sim p \to \sim q$, contrapositive _____. A conditional and its _____ are always both true or both false.

Example 2

Write the converse, inverse, and contrapositive of "If two angles form a linear pair, then they are supplementary." Decide whether each is true.

Biconditional: when a conditional and its converse are both true, combine them with the words _____.

Example 3

Write as a biconditional: "If two lines intersect to form a right angle, then they are perpendicular." (Its converse is also true.)

Part 2 · Partner practice

Partner practice

Write the contrapositive of “If m∠A = 30°, then ∠A is acute.” Is it true?